

Below is a pragmatic report design tailored to this repo. The central narrative should be: We implement and numerically study a quantum-channel cooling algorithm for preparing thermal and ground states of small transverse-field Ising systems, using exact diagonalization and superoperator spectral analysis to test convergence toward Gibbs states.

- Proposed Title

- “Numerical Study of Quantum-Channel Cooling for Thermal and Ground-State Preparation”
- Subtitle: “Exact-diagonalization benchmarks for the transverse-field Ising model”

- Report Structure

1. Abstract

- 150-250 words.
- State problem: preparing low-temperature/thermal quantum states.
- State method: averaged cooling channel with auxiliary qubit.
- State benchmark: transverse-field Ising model, exact Gibbs states.
- State main findings: convergence behavior, fixed-point quality, spectral gap/mixing-time dependence on parameters.

2. Introduction

- Motivation: ground-state and thermal-state preparation are central for quantum simulation.
- Explain why cooling channels are useful compared with direct variational/state-preparation approaches.
- Position this as an exploratory implementation for small systems, consistent with README.md:2.
- End with concrete research questions: Does the implemented averaged channel converge to the Gibbs state?
- How do β , h/J , α , σ , T , and τ affect convergence?
- Is convergence still good outside the weak-coupling regime suggested by theory?

3. Theoretical Background

- Quantum states and density matrices Define pure/mixed states, density matrix ρ , trace preservation, positivity.
- Thermal states Define Gibbs state:

$$\rho_\beta = \frac{e^{-\beta H}}{Z}, \quad Z = \text{Tr}(e^{-\beta H})$$

- Relate $\beta \rightarrow \infty$ to ground-state preparation.
- This is directly benchmarked in `ed.py:40`.
- Transverse-field Ising model Use:

$$H = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i$$

- Periodic boundary conditions are implemented in `cooling_channel.py:15` and `ed.py:15`.
- Discuss phases qualitatively: ferromagnetic regime, paramagnetic regime, critical region near $h/J \approx 1$ in the thermodynamic limit.
- Quantum channels Define completely positive trace-preserving maps:

$$\rho \mapsto \mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger$$

- Explain fixed points: ρ_* such that $E(\rho_*) = \rho_*$.
- Explain convergence via repeated application and spectral gap.
- Cooling-channel idea: System couples to one auxiliary qubit sampled from a thermal environment.
- Each step samples an interaction operator A and frequency ω ; see `cooling_channel.py:196`.
- The environment qubit is initialized thermally according to β and ω ; see `cooling_channel.py`.
- The joint evolution alternates interaction, system evolution, environment phase evolution, and interaction again; see `cooling_channel.py:107`.
- Superoperator representation: Vectorize density matrices and represent the channel as a matrix S .
- Explain Choi/Kraus construction.
- The repo constructs both Kraus and Choi forms in `superoperator.py:85` and `superoperator.py:100`.
- The averaged channel over operators/frequencies is built in `superoperator.py:122`.

4. Implementation

- Software architecture `cooling_channel.py`: circuit/channel construction.
- `ed.py`: exact diagonalization and target Gibbs/ground states.
- `superoperator.py`: averaged channel, spectrum, fixed-point analysis.
- `path_analysis.py`: direct repeated-channel simulations and metrics.
- `*_sweep.py`: parameter sweeps.
- Model: Small N , usually $N=4$ or $N=6$, due to exponential Hilbert-space scaling.
- Channel parameters: β : inverse temperature.
- α : system-environment coupling.
- σ : Gaussian time/frequency width.
- T : total interaction window.
- τ : Trotter/internal timestep.
- $\omega_{\max}$: maximum sampled bath frequency.
- `op_set`: sampled system coupling operators, e.g. XZ, XYZ; see `cooling_channel.py:39`.
- Numerical validation: Exact Gibbs target from diagonalization.
- Trace distance:

$$D(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1$$

- Fixed-point check implemented in `superoperator.py:193`.
- Spectral quantities extracted in `superoperator.py:148`.

5. Numerical Experiments

- Experiment 1: Direct cooling trajectory: Start from a simple initial state.
- Apply increasing numbers of cooling steps.
- Plot trace distance to Gibbs state versus number of steps.
- Use existing figures like `docu/distance_to_gibbs.png`.
- Experiment 2: Superoperator spectrum: Build averaged channel S .
- Plot eigenvalues in the complex plane.
- Identify eigenvalue near 1 and associated fixed point.
- Use Δ_2 as convergence proxy.
- Experiment 3: β sweep: Study lower-temperature behavior.
- Plot trace distance of fixed point to Gibbs state versus β .

- Plot spectral gap/mixing time versus β .
- Use existing data/superoperator_N4_beta_sweep_*.npz.
- Experiment 4: h/J sweep Sweep transverse field.
- Look for harder convergence near the TFIM critical region.
- Use superoperator_h_sweep.py.
- Experiment 5: Coupling/time-width regime Compare weak versus stronger coupling.
- This connects to the existing technical-doc claim that stronger coupling may converge faster than conservative guarantees suggest in docu/technical_docu.tex:77.

6. Results

- Organize by metrics, not by scripts: Fixed-point accuracy: $D(\rho_{\text{fix}}, \rho_\beta)$.
- Mixing time: iterations to reach chosen ϵ .
- Spectral gap proxy: distance between leading and subleading eigenvalues.
- Parameter sensitivity: $\beta, h/J, \alpha, \sigma, T, \tau$.
- Good core figures: Trace distance to Gibbs vs iterations.
- Spectrum of averaged superoperator.
- Fixed-point trace distance vs β .
- Mixing time vs h/J .
- Gap proxy vs h/J or β .

7. Discussion

- Interpret whether the cooling channel finds the correct thermal fixed point.
- Explain when convergence slows down.
- Discuss finite-size limitations.
- Be explicit that exact diagonalization limits this to small systems.
- Discuss discrepancy between theoretical weak-coupling assumptions and observed stronger-coupling performance.

8. Limitations

- Small system sizes only.
- Trotterization and quadrature errors.
- Random/operator sampling choices may bias convergence.
- Current work focuses on TFIM; impurity/DMFT milestone is planned but not yet central, per README.md:15.
- No hardware noise model yet, despite being listed as an open question in README.md:20.

9. Conclusion and Outlook

- Summarize implementation and numerical evidence.
- State whether the channel reliably approaches Gibbs states in tested regimes.
- Future work: Larger systems.
- Noise/NISQ simulation.
- Impurity model.
- Hybrid cooling + filtering.
- Resource estimation.